What Is an Option?

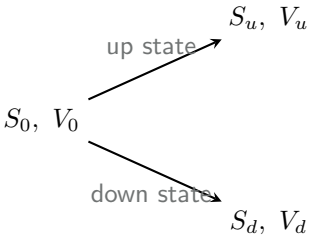
An option gives its holder the right, not the obligation, to transact in an underlying asset on specified terms.

- **Call:** right to buy at the strike price K.
- **Put:** right to sell at the strike price K.
- **Expiration T:** last date the right can be exercised.
- **Premium:** price paid today for the contract.

Central question

How much should someone pay today for a claim that provides a potentially uncertain payoff in the future?

An Arbitrary Financial Claim



Suppose a stock trades today at S_0. At time T, it can be S_u or S_d. A claim V written on the stock then pays V_u or V_d.

Replicating the Financial Claim

Rather than price the claim directly, build a portfolio with exactly the same payoff in every state.

Hold Δ shares of stock and borrow y dollars today at continuously compounded rate r.
Δ is the shares you have bought; y is the amount of the loan.

$$V_u = \Delta S_u - ye^{rT},$$
$$V_d = \Delta S_d - ye^{rT}.$$

Solving for Δ and the Loan

$$V_u - V_d = \Delta(S_u - S_d)$$

$$\therefore \qquad \Delta = \frac{V_u - V_d}{S_u - S_d}$$

$$y = e^{-rT}(\Delta S_d - V_d)$$

Δ is the shares you have bought. y is the amount of the loan.

No-Arbitrage: The Law of One Price

Portfolio lemma

If two portfolios have identical future payoffs in every state, then an arbitrage-free market must assign them the same price today.

$$V_0 = \Delta S_0 - y$$

A One-Step Call Option Example

$S_0 = \$100$, $S_u = \$120$, $S_d = \$80$, $K = \$100$, and $r = 0$. Therefore,

$$V_u = \max(120 - 100, 0) = \$20, \qquad V_d = \max(80 - 100, 0) = \$0,$$

$$\Delta = \frac{20 - 0}{120 - 80} = 0.5, \qquad y = 0.5(80) - 0 = \$40,$$

$$V_0 = 0.5(100) - 40 = \boxed{\$10}.$$

Up state

Half a share is worth $\$60$. Repay $\$40$.
Remaining payoff: $\$20$.

Down state

Half a share is worth $\$40$. Repay $\$40$.
Remaining payoff: $\$0$.

The $\$10$ premium is paid today; $\$20$ or $\$0$ is paid at expiration.

From Replication to Risk-Neutral Pricing

Insert the replicating portfolio into $V_0 = \Delta S_0 - y$:

$$\begin{aligned} V_0 &= \Delta S_0 - e^{-rT}(\Delta S_d - V_d) \\ &= e^{-rT}\left[qV_u + (1-q)V_d\right], \end{aligned}$$

where

$$q = \frac{S_0 e^{rT} - S_d}{S_u - S_d}.$$

Interpretation:

What Are Risk-Neutral Probabilities?

$$S_0 = e^{-rT}\left[qS_u + (1-q)S_d\right]$$

The weights q and $1-q$ make the stock's expected return equal to the risk-free rate.

Actual probabilities

Describe beliefs about what will happen in the world.

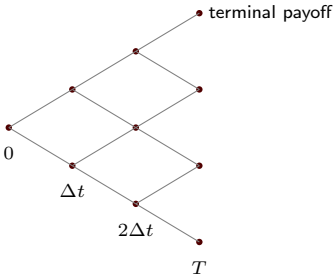
Risk-neutral probabilities

Are pricing weights consistent with traded-asset prices and no-arbitrage.

No-arbitrage condition

$S_d < S_0 e^{rT} < S_u$ ensures $0 < q < 1$. These weights arise from replication, not from forecasting the next stock move.

Extending to Multiple Time Steps



At every terminal node, the contract tells us the payoff. Then work backward:

$$V_t = e^{-r\Delta t}\left[qV^u_{t+\Delta t} + (1-q)V^d_{t+\Delta t}\right].$$

1. Calculate terminal payoffs.
2. Weight and discount adjacent values.
3. Repeat until time zero.

Terminal nodes are model-generated scenarios, not forecasts of realized prices.

From Binomial Trees to Black–Scholes

Discrete binomial model

Stock prices move up or down at individual time steps.

$\rightarrow$

Continuous-time model

Stock prices evolve continuously under a stochastic process.

More steps, smaller Δt

Under an appropriate parameterization, the binomial price converges to the Black–Scholes–Merton price as the number of steps grows.

Black–Scholes–Merton: Equation and Price

Continuous-time delta hedging and no-arbitrage lead to

$$\frac{\partial V}{\partial t} + \frac{1}{2}\sigma^2 S^2 \frac{\partial^2 V}{\partial S^2} + rS\frac{\partial V}{\partial S} - rV = 0.$$

For a European call with no dividends,

$$C = S_0 N(d_1) - K e^{-rT} N(d_2),$$

$$d_1 = \frac{\ln(S_0/K) + (r + \sigma^2/2)T}{\sigma\sqrt{T}}, \quad d_2 = d_1 - \sigma\sqrt{T}.$$

The Greeks: Sensitivities of Option Value

Delta: $\frac{\partial V}{\partial S}$

Sensitivity to stock price; locally, the shares needed to hedge the option.

Gamma: $\frac{\partial^2 V}{\partial S^2}$

Sensitivity of Delta to the stock price.

Theta

Sensitivity to time; its sign depends on whether time is calendar time or time to expiration.

Vega: $\frac{\partial V}{\partial \sigma}$

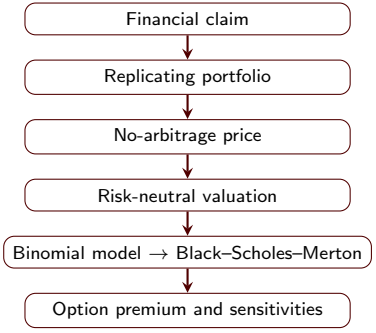
Sensitivity to volatility.

Rho: $\frac{\partial V}{\partial r}$

Sensitivity to the interest rate.

Connection: the Δ from the one-step replicating portfolio becomes a local, continuously adjusted hedge ratio.

Final Takeaway



Central message

Options pricing is not fundamentally about predicting where a stock will go. It is about determining what a financial claim should cost today, given its possible future payoffs and the prices of tradable assets.